

EXPERIMENT - 8

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8.1.1] Multiplication of Two Square Matrices

A) ALGORITHM

Step 1: Start

Step 2: Read the dimension n of the square matrices.

Step 3: Input the elements of first matrix A of size $n \times n$.

Step 4: Input the elements of second matrix B of size $n \times n$.

Step 5: Initialize a result matrix C of size $n \times n$ with all elements as 0.

Step 6: For $i = 0$ to $n-1$

 For $j = 0$ to $n-1$

 For $k = 0$ to $n-1$

$C[i][j] = C[i][j] + A[i][k] * B[k][j]$

Step 7: Display the resultant matrix C .

Step 8: Stop

B) PYTHON CODE

```
n = int(input("dimension: "))
print("first matrix:")
A = []
for i in range(n):
    r = list(map(int, input().split()))
    A.append(r)
print("second matrix:")
B = []
for i in range(n):
    r = list(map(int, input().split()))
    B.append(r)
result = []
for i in range(n):
    result.append([0]*n)
for i in range(n):
    for j in range(n):
        for k in range(n):
```

```

        result[i][j] += A[i][k] * B[k][j]

print("Resultant Matrix:")

for i in range(n):
    for j in range(n):
        if j == n - 1:
            print(result[i][j], end="")
        else:
            print(result[i][j], end=" ")

    print()

```

C) FLOWCHART



D) EXECUTION

CODETANTRA Home

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8.1.1. Multiplication of Two Square Matrices

01:37

Write a Python program to perform the multiplication of two square matrices. The program should take two matrices as input and output their product.

Input Format:

- The first line prints the "dimension:" and reads an integer n , representing the dimensions of the square matrices ($n \times n$).
- The second line prints: "first matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the first matrix.
- Next line prints: "second matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the second matrix.

Output Format:

- Print "Resultant Matrix:" followed by n lines, each containing n integers separated by spaces, representing the product of the two matrices.

Sample Test Cases

Debugger

Explorer

Multiplica...

```
1 # Type Content here...
2 n = int(input("dimension: "))
3 print("first matrix:")
4 A = []
5 for i in range(n):
6     r = list(map(int, input().split()))
7     A.append(r)
8
9 print("second matrix:")
10 B = []
11 for i in range(n):
```

Submit

Average time
0.029 s

Maximum time
0.043 s

29.25 ms

43.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 65 ms

Debug

Actual output

Expected output

dimension: 2

first matrix:

1 2

3 4

second matrix:

5 6

Terminal

Test cases

< Prev

Reset

Submit

Next >